

1. Introduction

AbilityJobs is a dedicated web-based employment portal connecting Persons with Disabilities (PWDs) with inclusive employers and NGOs. The platform addresses the critical gap in accessible, disability-centric job portals in India.

1.1 Objectives

- Build a WCAG 2.1 AA compliant, accessible job portal
- Enable multi-step registration capturing disability-specific information
- Allow employers to post disability-friendly jobs and review applications
- Integrate NGO support services for job seekers
- Deploy publicly on Render for live access

1.2 Scope

- Job Seeker registration, profile management, and application tracking
- Employer dashboard for job posting and candidate search
- NGO module for listing support services
- Client-side data storage via LocalStorage API
- Responsive design for mobile, tablet, and desktop

2. Problem Statement

Despite over 26 million persons with disabilities in India, only 36% of working-age PWDs are employed compared to 78% of non-disabled individuals. Key barriers include:

- Mainstream job portals lack screen reader support and accessibility features
- Employers are unaware of disability hiring quotas (3–5% under RPwD Act 2016)
- No centralized platform connecting PWDs, employers, and NGO support services
- 70% of PWDs report platform inadequacy as a primary employment barrier

Metric	Value
PWDs in India	26 million+
PWD employment rate	36%
Non-disabled employment rate	78%
Mandatory hiring quota	3–5%

3. Technology Stack

Technology	Purpose
HTML5	Semantic structure, ARIA labels, accessibility
CSS3 + Tailwind CSS v4	Responsive design, utility classes, themes
JavaScript ES6+	Business logic, DOM manipulation, validation
LocalStorage API	Client-side data persistence
FileReader API	Photo and resume file upload handling
Render (Hosting)	Free static site deployment with auto-CI/CD

3.1 Accessibility Standards

- WCAG 2.1 Level AA compliance
- Screen reader support (ARIA roles and labels)
- Full keyboard navigation
- High contrast mode, minimum 4.5:1 color contrast ratio

4. Key Features

4.1 For Job Seekers (PWDs)

- 3-step registration: Personal Info → Professional Info → Document Upload
- Profile strength meter with completion checklist
- 500+ job listings with disability-friendly filters
- Application tracking: Applied → Under Review → Shortlisted → Offer
- QR code profile sharing for networking events

4.2 For Employers

- Post jobs with accessibility requirements and accommodation details
- Search and filter PWD candidate database
- Track compliance with disability hiring quotas (3–5%)

4.3 For NGOs

- List support services, training programs, and counseling resources
- Connect directly with registered job seekers

5. Deployment on Render

AbilityJobs is deployed as a static site on Render (<https://render.com>) — a free cloud hosting platform with automatic CI/CD from GitHub.

5.1 Deployment Steps

1. Push project files to a GitHub repository
2. Sign up at <https://render.com> using GitHub account
3. Click "New +" → Select "Static Site"
4. Connect the AbilityJobs GitHub repository
5. Set Publish Directory to . (project root)
6. Click "Create Static Site" — auto-deploys in ~2 minutes
7. Access the live site at <https://abilityjobs.onrender.com>

5.2 Render Configuration (render.yaml)

```
services:
  - type: web
    name: abilityjobs
    env: static
    staticPublishPath: ./
```

5.3 Continuous Deployment

Every git push to the main branch triggers an automatic re-deployment on Render — no manual steps required after initial setup.

6. Impact & Future Scope

6.1 Benefits

Stakeholder	Benefit
Job Seekers (PWDs)	Accessible platform, career resources, equal opportunities
Employers	DEI compliance, qualified PWD candidates, tax incentives
NGOs	Centralized reach to job seekers, program visibility
Society	Reduced unemployment, social inclusion, breaks stigma

6.2 Future Enhancements

- AI-powered job matching based on disability type and skills
- Native mobile app (iOS & Android)
- Real-time chat and video interview integration
- Multi-language support (Hindi + regional languages)
- Blockchain-based credential verification
- Backend migration to cloud database for scalability

7. Conclusion

AbilityJobs demonstrates how accessible web technology can drive meaningful social impact. By connecting PWDs, inclusive employers, and support NGOs on a single platform, it addresses a critical employment gap and promotes equal opportunity.

The project applies HTML5, CSS3, Tailwind CSS, and JavaScript ES6+ with full WCAG 2.1 AA compliance and is publicly deployed on Render via an automated CI/CD pipeline.

"Where Ability Matters More Than Disability"

References

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